JAIN COLLEGE OF ENGINEERING & TECHNOLOGY

Hubli, Karnataka – 580029

Department of Software Engineering / Computer Science & Engineering

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DEVELOPMENT AND CLOUD DEPLOYMENT OF A MODERN PERSONAL PORTFOLIO, RESUME, AND BIO-DATA WEB APPLICATION

Web Technology Laboratory Assignment 1 Report

Submitted in partial fulfillment of the requirements for the evaluation of

Web Technology Laboratory (Assignment 1)

Bachelor of Engineering in Software Engineering

SUBMITTED BY:

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UNDER THE GUIDANCE OF:

Project Guide / Faculty:

Department of Computer Science & Engineering

Jain College of Engineering, Hubli

Karnataka, India

ACADEMIC YEAR: 2026 – 2027

JAIN COLLEGE OF ENGINEERING & TECHNOLOGY

Hubli, Karnataka

Department of Software Engineering

CERTIFICATE OF EVALUATION

This is to certify that the laboratory project work entitled **"Development and Cloud Deployment of a Modern Personal Portfolio, Resume, and Bio-Data Web Application"** is a bonafide work carried out by **Cynthia Alexander** in partial fulfillment for the evaluation of **Web Technology Laboratory (Assignment 1)** during the academic year 2026–2027.

The project has been reviewed, examined, and verified on the live production environment at **https://assignment-1-topaz-alpha.vercel.app** and is found to meet the prescribed academic curriculum standards.

Signature of Project Guide	**Signature of Head of Department**
Faculty Advisor	Department of Software Engineering
Internal Examiner	**External Examiner**

DECLARATION

I, **Cynthia Alexander**, student of Bachelor of Engineering at Jain College, Hubli, hereby declare that the project work entitled **"Development and Cloud Deployment of a Modern Personal Portfolio, Resume, and Bio-Data Web Application"** submitted to the Department of Software Engineering is a record of original development work undertaken by me under the guidance of our esteemed faculty members.

The code, styling, visual assets, documentation, and cloud deployment published at **https://assignment-1-topaz-alpha.vercel.app** have been developed in adherence to academic integrity and ethical guidelines.

Place: Hubli, Karnataka

Date: 07 November 2026

Cynthia Alexander

Candidate

ACKNOWLEDGEMENT

I would like to express my profound gratitude to our **Project Guide and Faculty Members** of the Department of Software Engineering, Jain College, for their invaluable guidance, encouragement, and insightful feedback throughout the duration of this web technology laboratory assignment.

I also extend my sincere appreciation to the **Head of the Department** and the **Principal** of Jain College of Engineering for providing the requisite laboratory infrastructure, computational facilities, and continuous academic support.

ABSTRACT

The rapid proliferation of web technologies has made interactive, responsive personal websites a standard requirement for computing professionals to present their academic credentials, research endeavors, and technical portfolios. This laboratory project encompasses the engineering, design enhancement, version-control integration, and cloud deployment of a multi-page personal web application for **Cynthia Alexander**.

Originating from baseline laboratory exercise specifications, the application was upgraded using clean semantic HTML5, modern CSS3 layout systems (Flexbox, CSS Grid, and custom design tokens), lightweight vanilla JavaScript for event handling, and pure Scalable Vector Graphics (SVG) icons to ensure high visual fidelity without platform-dependent emojis. The application was version-controlled using Git, published to GitHub, and hosted on the Vercel Edge Network via an automated Continuous Deployment (CD) pipeline.

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CHAPTER 1: INTRODUCTION & PROBLEM STATEMENT

1.1 Background & Motivation

As part of the academic curriculum for the Web Technology Laboratory course, students are tasked with understanding the fundamental trifecta of client-side web development: HyperText Markup Language (HTML) for document semantics, Cascading Style Sheets (CSS) for responsive layout presentation, and JavaScript for client-side interactivity.

While traditional laboratory exercises focus on isolated static HTML files, modern industry expectations require developers to demonstrate full-cycle competencies: semantic structure, responsive design across diverse viewports, version control discipline, and cloud hosting with automated continuous deployment.

1.2 Problem Statement & Objectives

The objective of Assignment 1 is to design and develop an authentic personal web application containing three core foundational components:

1. **Home Page:** Introductory landing page presenting candidate identity, navigation links, and interactive engagement.
2. **Bio-Data Page:** Structured tabular presentation of personal verification details, institutional declaration, and authentication block.
3. **Curriculum Vitae / Resume Page:** Academic record, technical proficiencies, project portfolio, achievements, and co-curricular interests.

In addition to satisfying the laboratory reference requirements, the project was extended to include a dedicated **About Me** module and an interactive **Contact Page**, styled under a modern design system and deployed live on the public cloud.

1.3 Project Deliverables & Metadata

Project Title	Personal Portfolio, Resume & Bio-Data Web Application
Candidate Name	Cynthia Alexander
Institution	Jain College of Engineering & Technology, Hubli
Live Vercel URL	**https://assignment-1-topaz-alpha.vercel.app**

GitHub Repository	**https://github.com/cynthiaalexandergeorge-create/assignment-1**
Deployment Branch	`main` (Production Pipeline)

CHAPTER 2: SYSTEM REQUIREMENTS & ARCHITECTURE

2.1 Hardware and Software Environment

CATEGORY	SPECIFICATION / TOOL USED
Operating System	Microsoft Windows 11 (64-bit)
Development Environment	Antigravity Agentic IDE / Visual Studio Code
Markup & Styling	HTML5, CSS3 (Vanilla, CSS Grid & Flexbox, Design Tokens)
Scripting Language	ECMAScript 6 (JavaScript Vanilla)
Vector Graphics	W3C Scalable Vector Graphics (SVG 1.1 / 2.0)
Version Control	Git Distributed Version Control System (v2.55+)
Remote Repository	GitHub Cloud (Git over HTTPS / Personal Access Token)
Hosting Platform	Vercel Global Edge Network (Automated Webhook CI/CD)
Target Browsers	Google Chrome, Microsoft Edge, Mozilla Firefox, Safari

2.2 Architectural Flow

The web application follows a clean **Single-Source Static Architecture (SSSA)** with client-side routing between five discrete HTML pages sharing a unified stylesheet (`style.css`) and interactive logic script (`script.js`).

```
[ Local Workspace (D:\assignment 1) ]
  ├── index.html        (Home Showcase & Hero Section)
  ├── about.html        (Dedicated Profile & Core Identity)
  ├── resume.html       (Education Table, Skills, Projects, Achievements)
  ├── biodata.html      (Personal Details Record & Declaration)
  ├── contact.html      (Contact Cards & Interactive Message Form)
  ├── style.css         (Design Tokens, Responsive Grid & Media Queries)
  ├── script.js         (DOM Event Listeners, Mobile Nav & Form Handler)
  └── assets/           (Vector Portraits & Photography Assets)
       ↓ (Git Commit & Branch: main)
[ GitHub Repository: cynthiaalexandergeorge-create/assignment-1 ]
       ↓ (Automated Continuous Deployment Webhook)
[ Vercel Global Edge CDN (assignment-1-topaz-alpha.vercel.app) ]
```

CHAPTER 3: WEB PAGE MODULES & IMPLEMENTATION

3.1 Home Module (index.html)

The home page is engineered as the central showcase portal:

- **Sticky Navigation Bar:** Features the brand monogram badge (`C`), desktop navigation links, mobile hamburger trigger, and a highlighted call-to-action button (`Contact Me`).
- **Hero Section:** Incorporates an electric-lime tag (`Hello I Am`), high-contrast heading (`I'm Cynthia`), candidate introduction, and action triggers leading to the resume and contact pages.
- **Rotating SVG Vector Badge:** Utilizes an SVG `<textPath>` curving smoothly along a circular radius (`SOFTWARE ENGINEERING • CYNTHIA •`) executing continuous 360° keyframe animation around a high-resolution portrait.
- **Obsidian Feature Ticker:** A dark marquee banner with lime sparkle icons presenting primary engineering specializations.

3.2 About Me Module (about.html)

Presents an authentic narrative of Cynthia Alexander's academic journey at Jain College, detailing her interest in distributed computing, machine learning, and clean software architecture. Includes structured capsule tags denoting core engineering attributes.

3.3 Curriculum Vitae / Resume Module (resume.html)

Presents academic qualifications, technical proficiencies, and projects in an evaluated tabular format:

DEGREE / LEVEL	INSTITUTION	PASSING YEAR	PERCENTAGE
B.E. (Software Engineering)	Jain College of Engineering, Hubli	2026	**85%**
PUC (12th Grade)	Adarsh College	2022	**80%**
SSLC (10th Grade)	Government School	2020	**90%**

Project Highlights Documented:

- *Student Management System:* Database-driven portal for recording student performance metrics.
- *Food Waste Management System:* Web system connecting surplus food donors with distribution networks.
- *AI in Federated Learning:* Decentralized machine learning framework prioritizing data privacy.

3.4 Bio-Data Module (biodata.html)

Formats official identification records: Full Name (*Cynthia Alexander*), Date of Birth (*01 January 2006*), Gender (*Female*), Nationality (*Indian*), Languages Known (*English, Kannada, Tamil, Hindi*), Residential Address (*Hubli, Karnataka*), formal legal declaration of truthfulness, and authentication signature block.

3.5 Contact Module (contact.html)

Provides direct communication channels (Email, Telephone, Location, and Student Internship Status) alongside an interactive client-side form validating inputs and providing immediate submission feedback.

CHAPTER 4: DESIGN TECHNIQUES & SVG ENGINEERING

4.1 Elimination of Casual Emojis in Favor of SVG Vectors

A critical design enhancement introduced during development was the complete elimination of casual platform emojis. Emojis render inconsistently across operating systems and diminish professional credibility.

All icons were re-engineered using **Scalable Vector Graphics (SVG)** with uniform 24×24 coordinate viewBox geometry, 2px stroke width, rounded line-caps, and dynamic color inheritance via `currentColor`:

```
<!-- Example: Professional SVG Email Icon -->
<svg width="18" height="18" viewBox="0 0 24 24" fill="none"
     stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round">
  <rect width="20" height="16" x="2" y="4" rx="2"/>
  <path d="m22 7-8.97 5.7a1.94 1.94 0 0 1-2.06 0L2 7"/>
</svg>
```

4.2 CSS Design Tokens & Palette Architecture

To ensure visual cohesion, CSS custom properties were implemented across all components:

- `--primary: #ff5e62` (Coral / Terracotta accent for buttons, active links, and brand icons).
- `--accent-lime: #ccff00` (High-visibility electric lime for badges and rotating highlights).
- `--dark-bg: #090a0f` (Obsidian black for ticker banners and footer elements).
- `--text-main: #0f172a` & `--text-muted: #64748b` (High-contrast WCAG 2.1 AA accessible typography).

4.3 Responsive Viewport Strategy

The stylesheet implements progressive enhancement with fluid media queries:

- **Desktop (> 900px):** Two-column CSS Grid layouts with generous spacing and visible horizontal navigation.

- **Tablet (650px – 900px):** Grid columns collapse to single-column centered layouts while maintaining card containment.
- **Mobile (< 650px):** Navigation transitions to a clean slide-down menu toggled via JavaScript, tables receive horizontal scrolling wrappers, and typography scales appropriately.

CHAPTER 5: DEVOPS PIPELINE, TESTING & CLOUD DEPLOYMENT

5.1 Version Control with Git

The repository was initialized locally using Git. A `.gitignore` configuration was deployed to exclude cache files. Commits followed standard semantic versioning practices documenting structural and stylistic milestones:

- `Initial commit: Personal Website structure (Home, Bio-data, Resume)`
- `Update personal details for Cynthia Alexander`
- `Add modern Home, Navigation, About Me, and Contact pages`
- `Replace casual emojis with professional SVG vector icons and photography`

5.2 GitHub and Vercel Continuous Deployment (CD)

The local repository was synchronized with the remote GitHub repository at `cynthiaalexandergeorge-create/assignment-1` on the `main` branch.

Vercel was integrated with the GitHub repository via automated deployment webhooks. Upon detecting any commit to the `main` branch, Vercel initiates an automated build, generates an optimized static deployment, assigns edge-cached SSL certificates, and updates the public production URL:

https://assignment-1-topaz-alpha.vercel.app

5.3 Production Verification Audit

Automated HTTP requests were executed against the live Vercel production endpoints to verify status codes and asset delivery:

PAGE MODULE	PUBLIC ENDPOINT ROUTE	HTTP STATUS	VERIFICATION STATE
Home	`https://assignment-1-topaz-alpha.vercel.app/`	**200 OK**	Verified live with all assets.

PAGE MODULE	PUBLIC ENDPOINT ROUTE	HTTP STATUS	VERIFICATION STATE
About Me	`https://assignment-1-topaz-alpha.vercel.app/about.html`	**200 OK**	Verified live with portrait.
Resume / CV	`https://assignment-1-topaz-alpha.vercel.app/resume.html`	**200 OK**	Verified live with qualifications table.
Bio-Data	`https://assignment-1-topaz-alpha.vercel.app/biodata.html`	**200 OK**	Verified live with declaration.
Contact	`https://assignment-1-topaz-alpha.vercel.app/contact.html`	**200 OK**	Verified live with interactive form.

CHAPTER 6: CONCLUSION, VERIFICATION & FUTURE SCOPE

6.1 Summary of Achievements

This laboratory project successfully achieved all pedagogical requirements specified in Assignment 1 while exceeding baseline criteria through professional UI/UX engineering:

1. Designed and implemented all required pages: **Home**, **Bio-Data**, and **Resume**, along with supplemental **About Me** and **Contact** modules.
2. Maintained complete accuracy with candidate personal credentials: Cynthia Alexander, Jain College (85%), Adarsh College (80%), Government School (90%), and active engineering projects.
3. Implemented industry-grade graphic standards by replacing emojis with crisp SVG vector icons and high-resolution photography.
4. Successfully integrated Git version control, GitHub remote repository synchronization, and automated continuous deployment on the Vercel Edge Network.

6.2 Future Enhancements

- **Dark/Light Theme Toggle:** Implementing CSS custom property switching using local storage persistence.
- **Dynamic Blog / Publications Engine:** Integrating a headless content management system (CMS) or Markdown parser for technical article publication.
- **Serverless Contact Form API:** Connecting the contact form to a serverless backend function (e.g., SendGrid / Resend API) for automated email forwarding.

6.3 Academic References & Repository Links

- **Production Website:** https://assignment-1-topaz-alpha.vercel.app
- **GitHub Repository:** https://github.com/cynthiaalexandergeorge-create/assignment-1
- **MDN Web Docs:** HTML5 Semantic Elements, CSS Grid, and Flexbox Specifications.
- **W3C Recommendations:** Scalable Vector Graphics (SVG) 2.0 Standards.
- **Vercel Documentation:** Continuous Deployment and Static Site Hosting Guidelines.

Faculty Guide Approval:

Submitted by:

Signature & Seal

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